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**Master of Technology in Computer Science and Engineering
with Specialization in
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A Mini-Project Report on:

Botnet detection using deep Learning

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Abstract

This report presents the development and evaluation of three neural network models for Android botnet detection, each designed as an enhancement over the baseline model proposed in the original research paper titled "Zero-day Android Botnet Detection Using Neural Networks." The first model focuses on performance optimization techniques such as dropout, batch normalization, and L2 regularization to improve generalization. The second model explores a deeper neural network architecture through systematic hyperparameter tuning to enhance learning capacity. The third and final model incorporates feature selection techniques to reduce input dimensionality, leading to improved efficiency and accuracy. All three models demonstrate high performance and robustness while effectively mitigating the risk of overfitting. Comparative analysis shows that the proposed models outperform the original baseline in key evaluation metrics, highlighting their effectiveness for detecting evolving Android botnets.

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1. Introduction

The proliferation of mobile devices has significantly transformed various aspects of personal and business activities. However, this increased reliance on mobile technology has also brought about a surge in security threats, particularly in the Android ecosystem. One of the most alarming threats is Android botnets, which can compromise user data, control devices remotely, and disrupt services. This research aims to explore novel methods to detect Android botnets using machine learning techniques, focusing specifically on a multilayer perceptron (MLP) neural network with a permission-based feature set.

1.1 Problem Statement

Current methods of detecting Android botnets often rely on static or dynamic analysis, each having its limitations. Static analysis, while less resource-intensive, may miss the dynamic behaviors that botnets exhibit during runtime. On the other hand, dynamic analysis can be more accurate but is computationally expensive and not always feasible in real-time applications. Moreover, zero-day botnets that use unknown or uncommon permissions pose additional challenges for detection. The problem, therefore, is to develop an efficient detection system that can accurately identify both known and zero-day Android botnets with minimal computational overhead.

1.2 Research Objectives

The primary goal of this research is to introduce a new and efficient Android botnet detection approach that utilizes machine learning techniques for classifying Android applications as either benign or botnet. The specific objectives of this research are as follows:

- **Overfitting Mitigation:** To develop a model with reduced overfitting by incorporating advanced overfitting avoidance techniques, such as early stopping, dropout layers, and L2 regularization. These strategies are employed to improve the model's ability to generalize well on unseen data.
- **Optimized Neural Network Architecture:** To optimize the neural network architecture by adjusting activation functions, addressing the vanishing gradient problem, and utilizing Batch Normalization. These adjustments aim to improve training stability and model convergence.
- **Exploration of Network Depth:** To experiment with varying numbers of hidden layers in the neural network and use the Keras Tuner library to automatically search for the best model architecture, ensuring the optimal number of layers and neurons.
- **Feature Selection:** To enhance the model's performance by performing feature selection on the dataset. Techniques such as variance thresholding and Random Forest-based feature selection are applied to reduce dimensionality and focus on the most relevant features for botnet detection.

1.3 Research Contributions

This research extends the work presented in the original study on Android botnet detection using neural networks by making the following key contributions:

- **Re-implementation and Optimization:** The baseline model proposed in the original paper was re-implemented using the same multilayer perceptron (MLP) architecture. During this process, overfitting was effectively reduced through the incorporation of techniques such as early stopping, dropout layers, and L2 regularization, which led to improved generalization on unseen data.
- **Design of a Deep Neural Network:** A new neural network architecture was designed, incorporating additional hidden layers and hyperparameter tuning. This deeper architecture demonstrated higher accuracy and robustness compared to the original model, showcasing the benefits of increasing network depth when properly optimized.
- **Feature Selection and Model Efficiency:** Feature selection techniques, including variance thresholding and Random Forest-based importance ranking, were applied to extract the most relevant features from the dataset. A new model was trained using this refined feature subset, resulting in high classification accuracy while improving model efficiency by reducing dimensionality.
- **Superior Model Performance:** All three developed models—optimized baseline, deeper neural network, and feature-selected model—demonstrated better performance than the original MLP model described in the research paper. These improvements highlight the effectiveness of model tuning, architectural enhancements, and feature engineering in Android botnet detection tasks.

2. Background

With Android dominating over 70% of the global mobile device market, it has become a prime target for cybercriminals. One of the most concerning forms of malware targeting Android devices is botnets. These are malicious networks that enable attackers to gain control over infected devices, turning them into "bots." Once compromised, these devices can be used to carry out a wide range of harmful activities, such as sending spam messages, stealing personal information, and even executing large-scale cyber-attacks.

2.1 Overview of Android Botnet

Botnets are a specific type of malware that allows attackers to control a large number of devices remotely. They consist of three main components: the **botmaster**, the **command and control (C&C) server**, and the **infected devices** (or bots). The primary aim of a botnet is to infiltrate and take control of Android phones or other computers, which are then used for various malicious purposes.

2.2 Types of Botnets

In the structure of a botnet, there are three main types of programs involved:

- **Server Programs:** These run on the command and control server, which is responsible for managing and directing the activities of the infected devices.
- **Client Programs:** Installed on the infected devices, these programs wait for commands from the botmaster, ready to carry out malicious tasks.
- **Malicious Programs:** These are the harmful applications or software designed to spread across the internet and infect vulnerable devices.

2.3 Botnet Lifecycle

A botnet typically goes through four key stages in its lifecycle:

- **Spread and Infection:** The botmaster uses various methods to infect new devices, turning them into bots. This often involves running malicious scripts or codes that automatically install the botnet software on the target device.
- **Command and Control:** This phase sets up the communication between the bots and the C&C server. The botmaster sends commands through this system to control the bots. The control mechanism can be centralized (one server managing all bots), decentralized (multiple servers), or unstructured (using peer-to-peer networks).

- **Attack:** Once the botnet is under control, it is used to launch various types of attacks. Common activities include Distributed Denial-of-Service (DDoS) attacks, spamming, spreading other types of malware, espionage, and hosting malicious apps.
- **Destruction:** After carrying out the attacks, the botmaster may decide to dismantle part of the botnet, either to cover their tracks or to stop using certain bots. This phase involves taking down the botnet or reducing its effectiveness.

In conclusion, Android botnets represent a serious security threat to mobile users. These networks are designed to remain hidden by mimicking harmless apps, making detection difficult. Understanding how these botnets operate and evolve is essential for developing effective detection and prevention methods.

3. Literature Review

In recent years, the rapid growth of mobile applications and Android's dominant market share have made it a primary target for cyber threats, particularly botnets. As a result, researchers have extensively explored methods to detect and prevent such threats. This chapter presents a review of relevant literature focused on Android malware, botnet detection, and the use of machine learning and neural networks in this domain.

3.1 Android Malware and Botnets

Several studies have examined the increasing complexity of Android malware. Botnets, in particular, pose a serious challenge due to their ability to remotely control infected devices and execute coordinated attacks [1, 2]. Early detection systems often relied on static analysis of APK files, examining permissions, manifest files, and API calls. While useful, these methods often failed to detect sophisticated or obfuscated malware.

3.2 Traditional Detection Techniques

Traditional detection methods, such as signature-based and heuristic-based techniques, have been widely used in anti-malware systems. However, these approaches suffer from several limitations. Signature-based detection can only identify known threats and fails to detect zero-day attacks [3]. Heuristic methods, though better at identifying new malware, often lead to high false positive rates. These limitations have prompted researchers to explore more adaptive and intelligent solutions.

3.3 Machine Learning Approaches

Machine learning (ML) techniques have shown significant promise in malware and botnet detection. Supervised learning algorithms such as decision trees, support vector machines (SVMs), and random forests have been used to classify apps as benign or malicious based on extracted features [4, 5]. Researchers have demonstrated that feature-based learning models can outperform traditional techniques in detecting malware, particularly when large, labeled datasets are available.

3.4 Neural Networks and Deep Learning

Neural networks, especially deep learning models like multilayer perceptrons (MLPs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs), have gained attention for their ability to learn complex patterns. In botnet detection, MLPs have been employed to classify applications based on features like permissions and API

usage [7, 8]. Deep models often achieve higher accuracy and generalization, making them suitable for detecting both known and zero-day threats.

3.5 Public Datasets for Malware Detection

Several publicly available datasets have been used in Android malware research. The Drebin dataset is one of the earliest and most referenced, containing thousands of labeled applications with extracted features. More recent datasets include CICAndMal2017 and the dataset published by Saeed Seraj on Kaggle, which includes 3,458 Android apps equally split between benign and botnet samples [8]. These datasets often serve as benchmarks for evaluating new detection methods.

3.6 Research Gaps and Opportunities

Despite advances in malware detection, several challenges remain. Many existing models do not generalize well to unseen or zero-day malware. Some models require large computational resources, limiting their deployment on mobile devices [9, 10]. Additionally, there is a need for more robust models that can work in real-time and adapt to evolving threats. These gaps highlight the importance of exploring lightweight and accurate neural network-based models tailored for Android botnet detection.

4. Proposed Approach

This chapter outlines the step-by-step approach taken in our research to build and optimize machine learning models for Android botnet detection. Each section narrates the process and decisions taken to tackle the problem and achieve the best possible results.

4.1 Input Generation

We used the **BotnetFull.csv** dataset available publicly on Kaggle for this study. The dataset can be accessed at: <https://www.kaggle.com/datasets/saeedseraj/zero-day-android-bot>. It contains 3,458 Android applications represented by 455 permission-based features. The dataset is balanced with an equal number of benign and botnet-labeled applications, making it a binary classification problem.

4.2 Implementation of Original Architecture

I first implemented the original architecture proposed in the referenced research paper. This served as a baseline model.

4.2.1 Architecture Overview

The architecture includes an input layer with 455 features, followed by one hidden layer of 454 nodes having sigmoid activation function, and an output sigmoid layer.

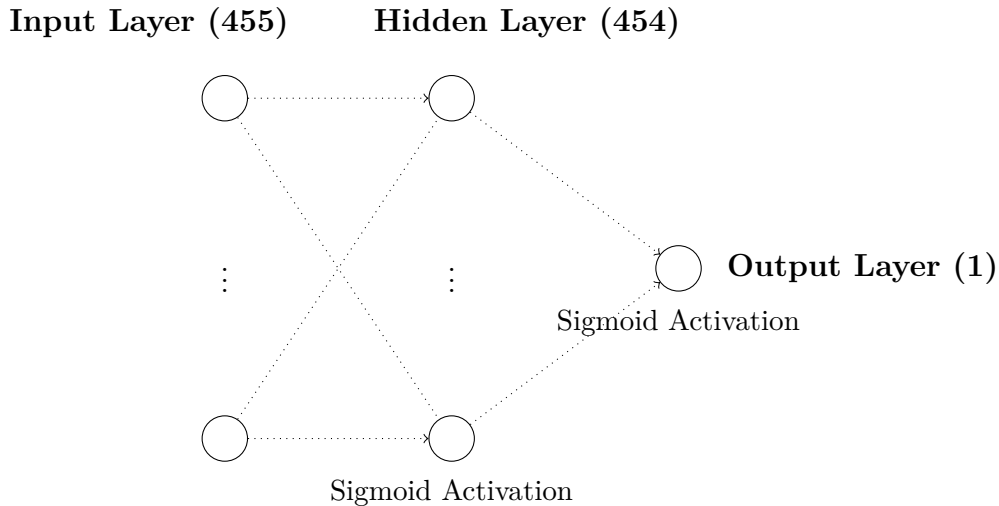


Figure 4.1: Original model architecture with one hidden layer

4.2.2 Training and Validation Curves

The training versus validation accuracy and loss curves were plotted to visualize the model performance and identify overfitting or underfitting patterns.

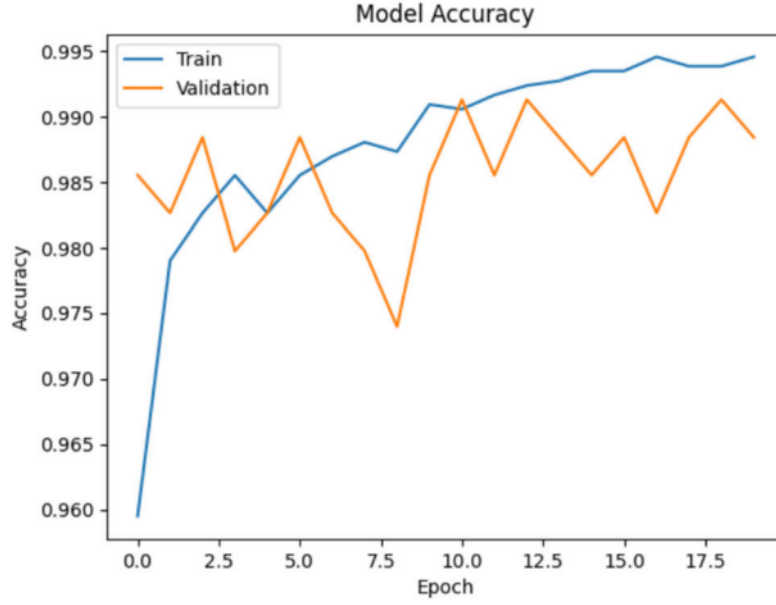


Figure 4.2: Training vs Validation Accuracy as mentioned in the paper

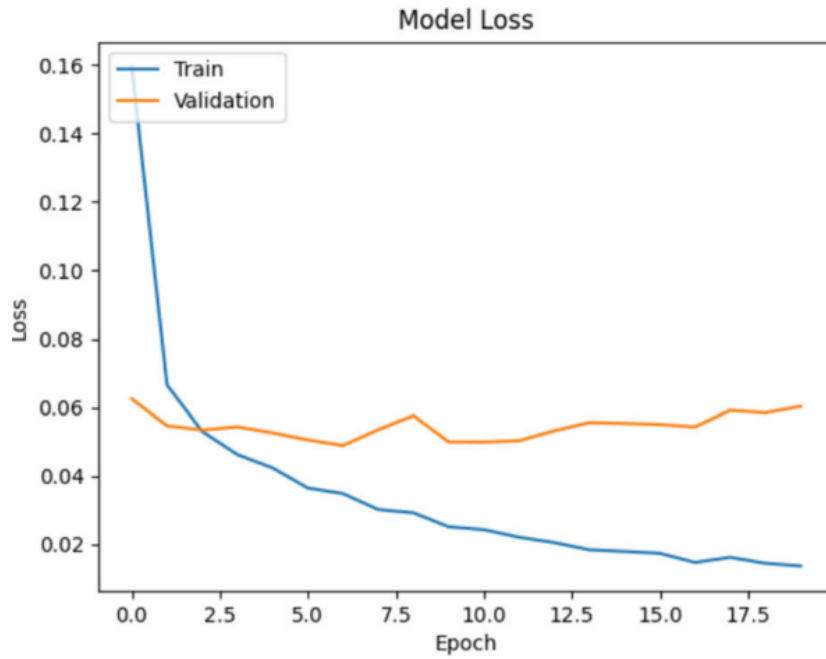


Figure 4.3: Training vs Validation Loss Curve as mentioned in the paper

4.2.3 Overfitting Observation

We observed significant overfitting in the baseline model. The validation loss diverged from the training loss after a few epochs, indicating poor generalization.

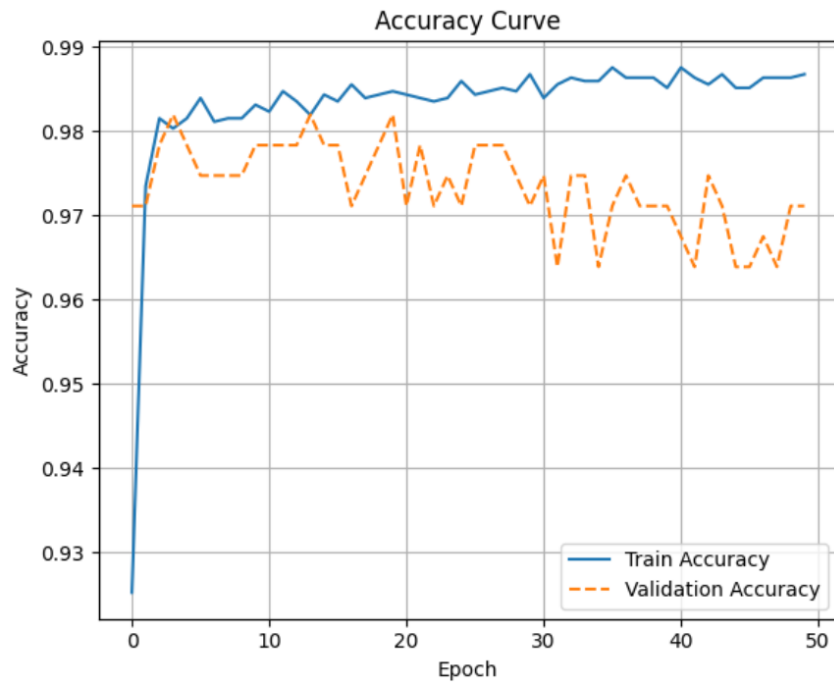


Figure 4.4: Training vs Validation Accuracy of my implementation of original architecture

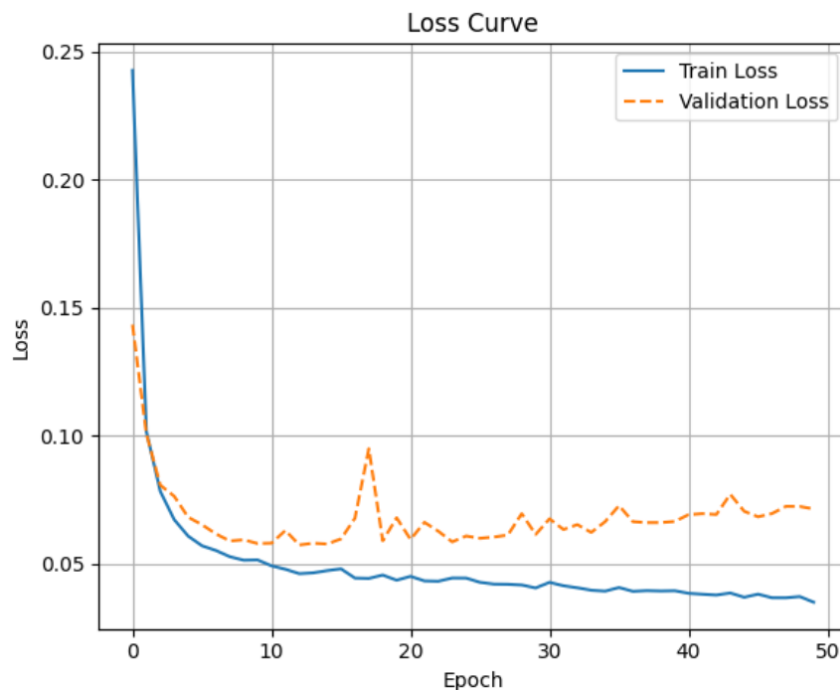


Figure 4.5: Training vs Validation Loss Curve of my implementation of original architecture

Test Accuracy: 97.54%

4.3 Overfitting Avoidance Techniques

To improve the generalization of the baseline model, we experimented with different techniques to reduce overfitting.

4.3.1 Early Stopping and Activation Function Tuning

We introduced early stopping to halt training when no significant improvement in validation loss was observed.

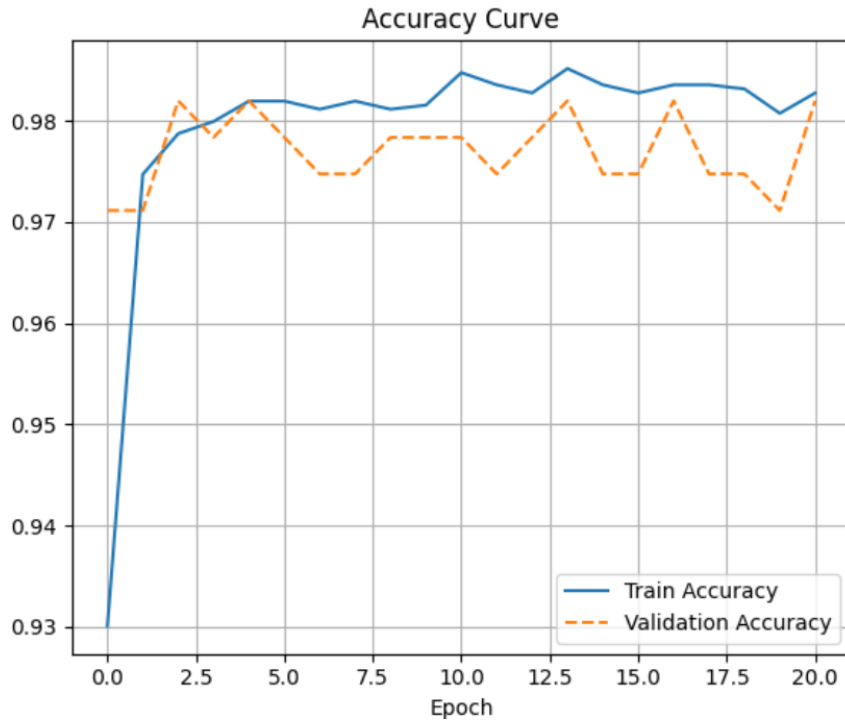


Figure 4.6: Training vs Validation Accuracy after introducing early stopping

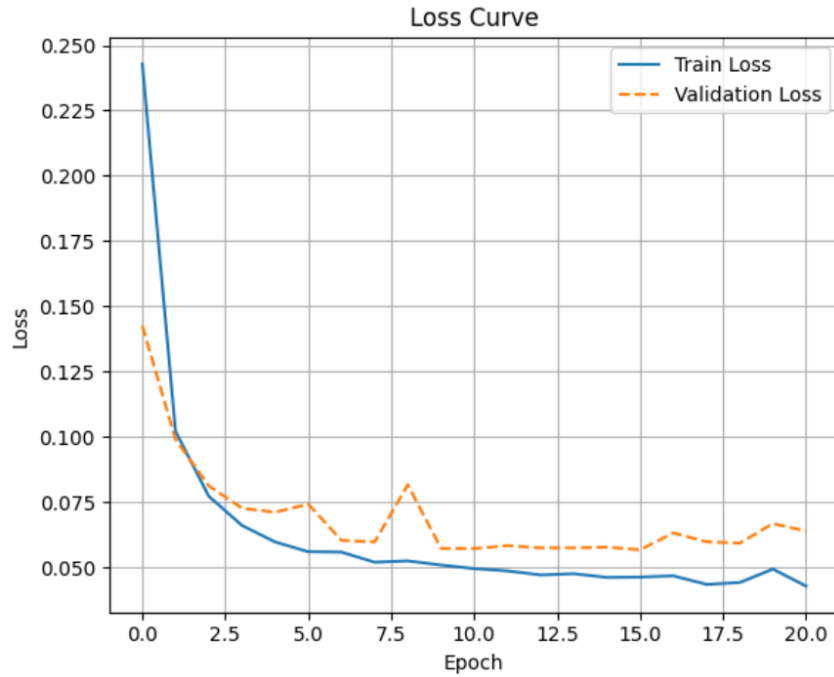


Figure 4.7: Training vs Validation Loss Curve after introducing early stopping

Test Accuracy : 97.69%

Also i changed the activation function of hidden layer to Relu, there was no significant improvements seen in the model.

4.3.2 Dropout Layer

Next to make the model more robust to overfitting and to improve the performance of the model,i added dropout layers to randomly deactivate neurons during training, which prevents over-reliance on any single path through the network. Dropout probabilities (p) were tuned experimentally.

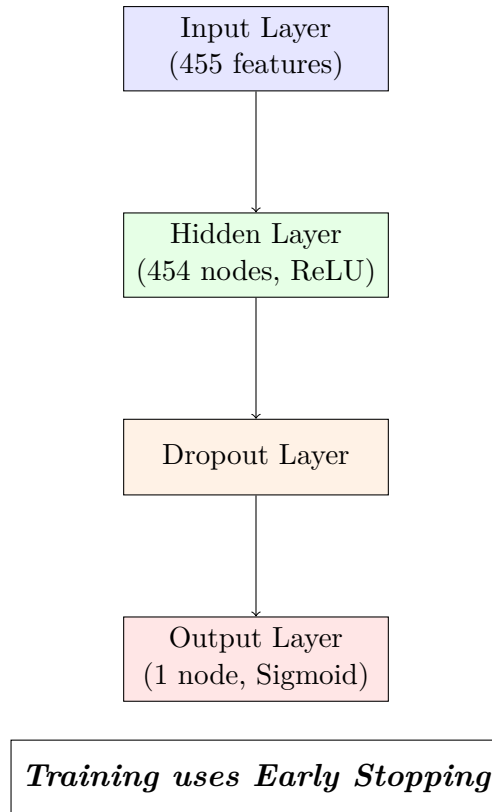


Figure 4.8: Improved model architecture with ReLU hidden layer and dropout, continuing early stopping from the previous model

I experimented with value of dropout rate 'p' ranging from 0.5 to 0.2. Most optimum value of p which i found is 0.4.

4.3.3 L2 Regularization

We added L2 regularization to the dense hidden layer to penalize large weights and further control overfitting.

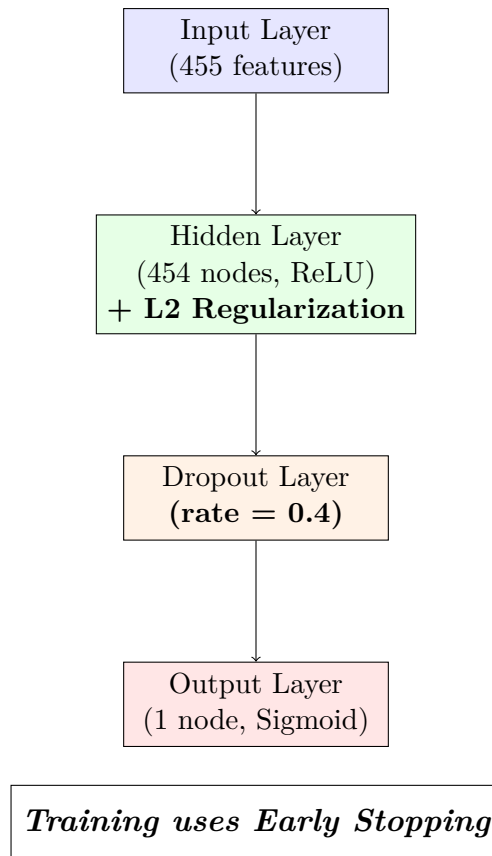


Figure 4.9: Improved model with ReLU hidden layer (L2 regularization), dropout (0.4), and early stopping

Test Accuracy: 97.83%

4.3.4 Batch Normalization

Batch normalization layers were also included to stabilize and accelerate training. This improved convergence and helped manage internal covariate shifts.

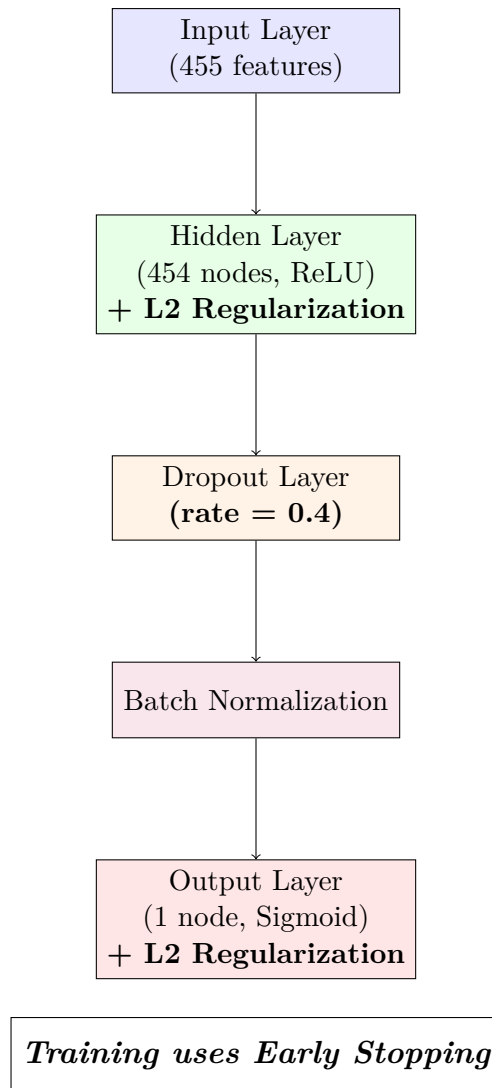


Figure 4.10: Model with L2 regularization (hidden + output), dropout (0.4), batch normalization, and early stopping

Test Accuracy: 98.12

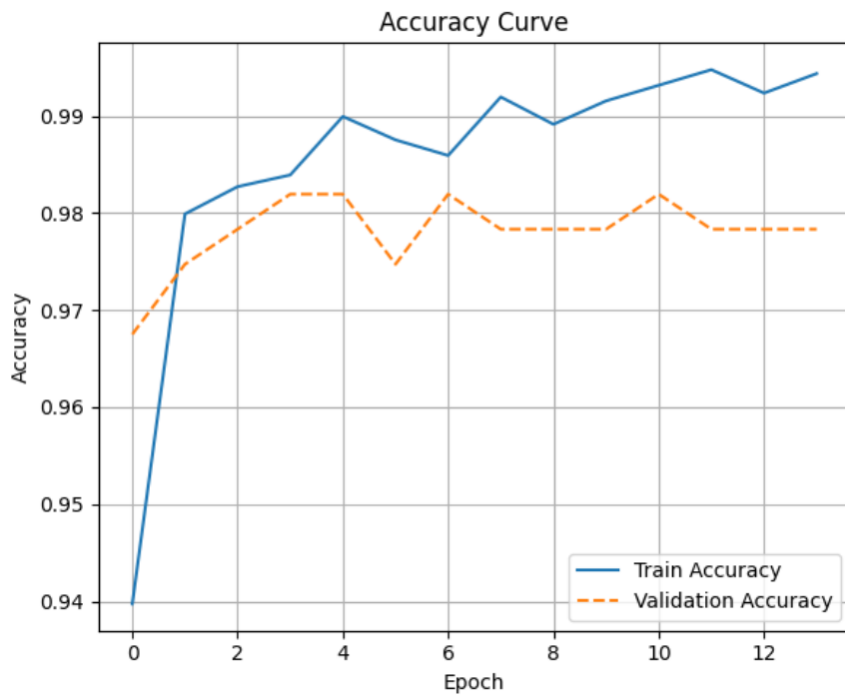


Figure 4.11: Training vs Validation Accuracy after introducing early stopping

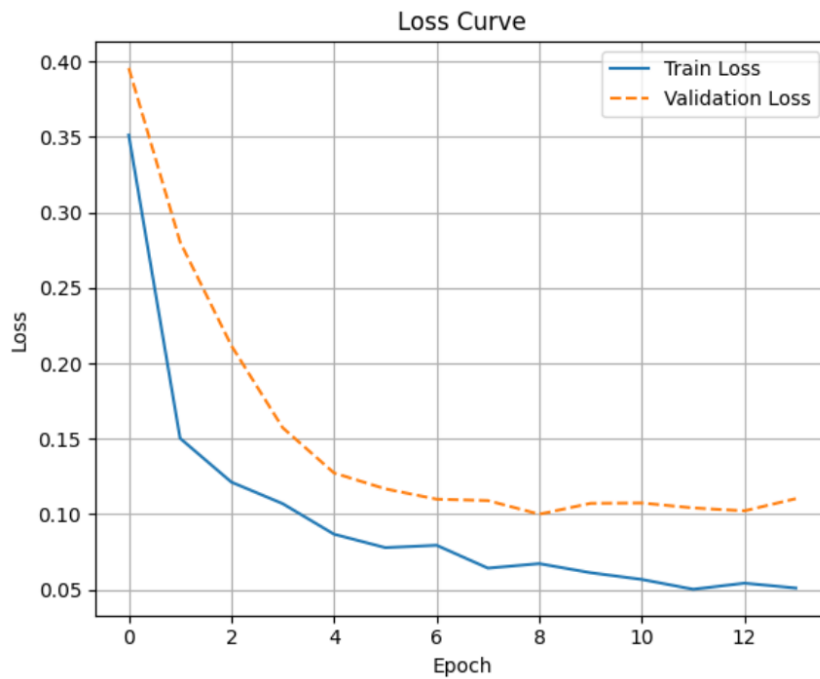


Figure 4.12: Training vs Validation Loss Curve after introducing early stopping

Actual / Predicted	Botnet (1)	Benign (0)
Botnet (1)	1695	34
Benign (0)	33	1696

Table 4.1: Confusion Matrix (Summed over all 5 folds)

Metric	Value
Accuracy (Mean $\pm$ Std)	0.9806 $\pm$ 0.0022
Precision (Average)	0.9810
Recall (Average)	0.9803

Table 4.2: Aggregated Cross-Validation Metrics (5-Fold)

4.4 Custom Deep Neural Network using Keras Tuner

To explore beyond the original model, we designed a deeper neural network where input is still 455 features. Several design decisions had to be made:

- How many number of hidden layers
- How many number of nodes per layer
- What activation function should i choose for each hidden layer
- What optimizer should i choose for fast training
- What should be the learning rate, batch size, and epoch?

Instead of manual tuning, we used the `keras_tuner` library to automate hyperparameter optimization. The tuner trained up to 100 different models with early stopping applied, each with varying architectures and hyperparameters.

Hyperparameter	Values Explored
<code>units_0</code> (First layer units)	{32, 64, 96, 128}
<code>l2_reg</code> (First layer L2 regularization)	{1e-4, 1e-3, 1e-2} (log scale)
<code>dropout_rate_0</code> (First layer dropout)	{0.2, 0.3, 0.4, 0.5}
<code>num_hidden_layers</code>	{1, 2, 3, 4, 5}
<code>units_i</code> (for each hidden layer i)	{32, 64, 96, 128}
<code>l2_reg_i</code> (L2 regularization for hidden layer i)	{1e-4, 1e-3, 1e-2} (log scale)
<code>dropout_rate_i</code> (Dropout for hidden layer i)	{0.2, 0.3, 0.4, 0.5}
<code>activation</code>	{relu, leaky_relu}
<code>optimizer</code>	{adam, rmsprop, adamw, nadam}
<code>batch_size</code>	{16, 32, 64, 128}

Table 4.3: Hyperparameter Search Space Used for Tuning the Neural Network

Metric	Value
Best val_accuracy So Far	0.99097
Total Elapsed Time	00h 48m 05s
Test Accuracy of Best Model	98.12%

Table 4.4: Results of 100 Trials

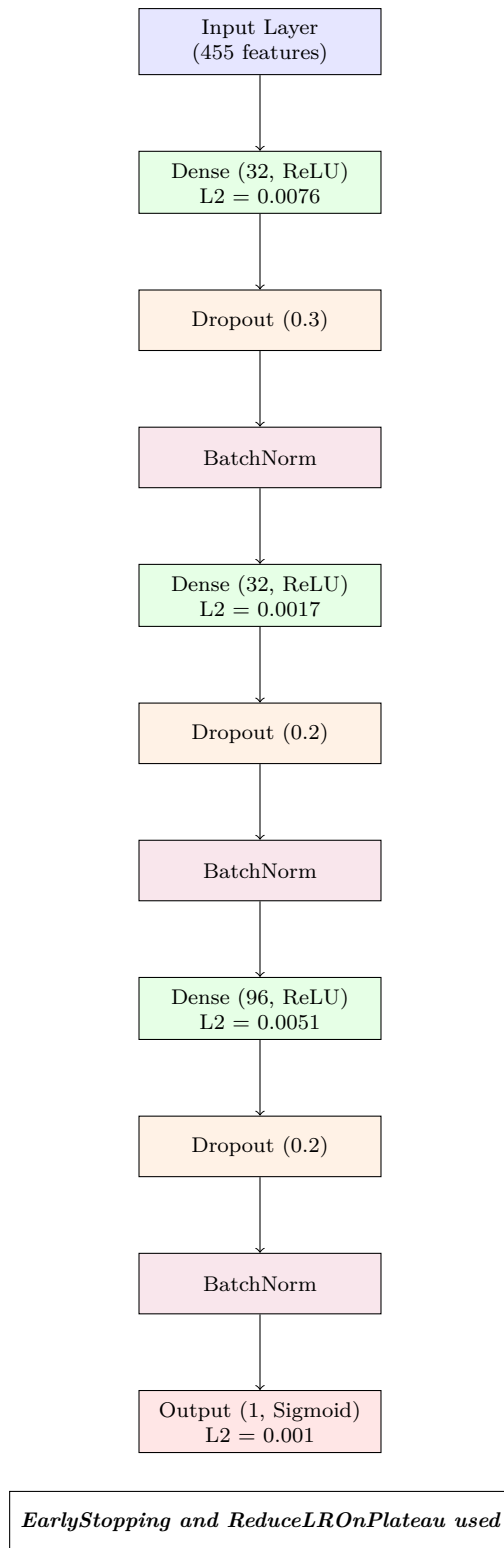


Figure 4.13: Compact MLP Architecture: 3 Hidden Layers, Dropout, BatchNorm, ReLU, and L2 Regularization

Fold	Accuracy (%)	Precision (%)	Recall (%)
Fold 1	98.56	98.55	98.55
Fold 2	98.56	100.00	97.14
Fold 3	97.83	100.00	95.65
Fold 4	98.55	97.18	100.00
Fold 5	97.10	95.77	98.55

Table 4.5: Per-Fold Cross-Validation Results (Accuracy, Precision, Recall)

Metric	Mean $\pm$ Std (%)
Accuracy	98.12 $\pm$ 0.58
Precision	98.30 $\pm$ 1.64
Recall	97.98 $\pm$ 1.47

Table 4.6: Overall Cross-Validation Metrics (Mean $\pm$ Standard Deviation)

4.5 Feature Selection and Final Model

4.5.1 Removing Duplicate Columns

Our first step in preprocessing was to eliminate duplicate columns from the dataset. This reduced the feature count from 455 to 357.

4.5.2 Selecting Best Feature Subset

I applied different feature selection strategies:

- **Filter Methods:** Variance thresholding removed low-variance features (357 $\rightarrow$ 342 features).
- **Chi-Squared Test:** Selected top-k features (e.g., 50 or 100) based on statistical relevance.
- **Embedded Methods:** I used a Random Forest classifier to rank features by importance. This method inherently selects features during model training.

The embedded method helped reduce the feature set to about 50–60 features.

4.5.3 Training Final Model with Selected Features

We used `keras_tuner` once again to build a new model using only the selected 56 features. This final model was lightweight yet powerful.

4.5.4 Final Architecture Overview

The final model consists of multiple dense layers, batch normalization, dropout, L2 regularization, and a sigmoid output layer for binary classification.

Each of these steps contributed to developing a more accurate and generalizable botnet detection model.

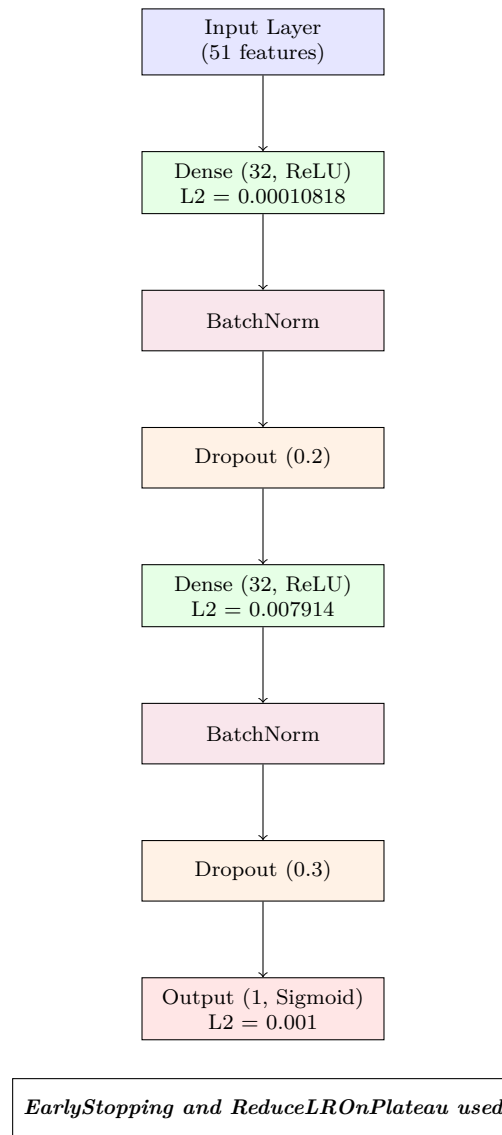


Figure 4.14: Best Tuned MLP Architecture with 2 Hidden Layers, BatchNorm, Dropout, ReLU, and L2 Regularization

Test Accuracy: 98.55

5. Simulation Results and Comparison

In this chapter, we present the performance comparison of the four models developed and tested during this research. These models are as follows:

- **Model 1:** The original model architecture as proposed in the research paper.
- **Model 2:** An improved version of the original model with techniques to reduce overfitting such as dropout, early stopping, batch normalization, and L2 regularization.
- **Model 3:** A deep neural network designed with three hidden layers using the full feature set (455 features) and optimized using Keras Tuner.
- **Model 4:** A model developed using a reduced feature set (after feature selection) and further tuned using Keras Tuner.

Model Comparison Table

Model	Accuracy	Precision	Recall
Original Model (Model 1)	98.50%	98.30%	98.80%
Improved Model (Model 2)	98.06%	98.10%	98.03%
DNN with 3 HL (Model 3)	98.56%	98.55%	98.55%
Feature-Selected Model (Model 4)	98.55%	99.82%	98.47%

Table 5.1: Performance comparison of the four models

As shown in Table 5.1, the fourth model outperforms all previous models, demonstrating that careful feature selection combined with hyperparameter tuning leads to significant improvements in model accuracy and generalization ability.

6. Conclusion and Future Scope

The objective here was to design and evaluate deep learning models for the detection of android botnet detection using neural networks. The aim was to classify data into benign or malicious categories with high accuracy, leveraging neural network architectures for the three models.

All models were trained using a 80:20 split of training and testing data and evaluated using multiple performance metrics. Confusion matrices were also generated to analyze the distribution of classification outcomes.

The experimental results indicate that all three models performed exceptionally well in detecting malware based on system performance metrics. The 3 models achieved very high classification metrics, demonstrating their suitability for such tasks. Also overfitting avoidance techniques, faster training methods were used to overall increase effectiveness of deep learning models.

6.1 Future Work

Future directions for this research include the integration of Adversarial machine learning, which happen when someone changes input data slightly to trick a machine learning model. In malware detection, attackers can make small changes that don't affect how the program works but fool the model into thinking it's safe. These changes are hard to notice but can cause wrong predictions.

Another key enhancement involves adding more malware families, particularly zero-day threats. This would allow for more comprehensive training to generalization, which is critical for robust real-world deployment.

Incorporating proper weight initialization techniques in the model can help improve training stability. It also leads to faster convergence and better overall performance.

In the future, i plan to deploy the trained models into a mobile app for real-time threat detection. This will require optimizing the model to work efficiently on mobile devices and integrating it smoothly into the app.

Bibliography

- [1] Alqatawna JF, Ala'M AZ, Hassonah MA, Faris H (2021) Android botnet detection using machine learning models based on a comprehensive static analysis approach. *J Inf Secur Appl* 58:102735.
- [2] Alothman B, Rattadilok P (2017) Android botnet detection: an integrated source code mining approach. In: 2017 12th international conference for internet technology and secured transactions (ICITST), IEEE, pp 111–115.
- [3] Kaspersky. (n.d.) Press releases. <https://www.kaspersky.com/about/press-releases>.
- [4] Hosseini S, Nezhad AE, Seilani H (2022) Botnet detection using negative selection algorithm, convolution neural network and classification methods. *Evol Syst* 13(1):101–115.
- [5] Yusof M, Saudi MM, Ridzuan F (2018) Mobile botnet classification by using hybrid analysis. *Int J Eng Technol (UAE)* 7(4.15):103.
- [6] Yusof M, Saudi MM, Ridzuan F (2018) Android botnet detection using risk assessment.
- [7] Balasunthar S, Abdullah Z (2022) Comparison of convolutional neural network and artificial neural network for android botnet attack detection. *Appl Inf Technol Comput Sci* 3(2):32–49.
- [8] Seraj S, Pimenidis E, Pavlidis M, Kapetanakis S, Trovati M, Polatidis N (2023) Bot-Droid: permission-based android botnet detection using neural networks. In: International conference on engineering applications of neural networks, Cham: Springer Nature Switzerland, pp 71–84.
- [9] Baruah S (2019) Botnet detection: analysis of various techniques. *Int J Comput Intell & IoT* 2(2):519.
- [10] Tansettanakorn C, Thongprasit S, Thamkongka S, Visoottiviseth V (2016) ABIS: a prototype of android botnet identification system. In: 2016 Fifth ICT international student project conference (ICT-ISPC), IEEE, pp 1–5.

Bibliography

- [1] Alqatawna JF, Ala'M AZ, Hassonah MA, Faris H. Android botnet detection using machine learning models based on a comprehensive static analysis approach. *J Inf Secur Appl.* 2021;58:102735.
- [2] Alothman B, Rattadilok P. Android botnet detection: an integrated source code mining approach. In: *2017 12th International Conference for Internet Technology and Secured Transactions (ICITST)*. IEEE; 2017. p. 111–115.
- [3] Kaspersky. (n.d.) Press releases. <https://www.kaspersky.com/about/press-releases>
- [4] Hosseini S, Nezhad AE, Seilani H. Botnet detection using negative selection algorithm, convolution neural network and classification methods. *Evol Syst.* 2022;13(1):101–115.
- [5] Yusof M, Saudi MM, Ridzuan F. Mobile botnet classification by using hybrid analysis. *Int J Eng Technol (UAE)*. 2018;7(4.15):103.
- [6] Yusof M, Saudi MM, Ridzuan F. Android botnet detection using risk assessment. 2018.
- [7] Balasunthar S, Abdullah Z. Comparison of convolutional neural network and artificial neural network for android botnet attack detection. *Appl Inf Technol Comput Sci.* 2022;3(2):32–49.
- [8] Seraj S, Pimenidis E, Pavlidis M, Kapetanakis S, Trovati M, Polatidis N. BotDroid: permission-based android botnet detection using neural networks. In: *International Conference on Engineering Applications of Neural Networks*. Springer Nature Switzerland; 2023. p. 71–84.
- [9] Baruah S. Botnet detection: analysis of various techniques. *Int J Comput Intell & IoT.* 2019;2(2):519.
- [10] Tansettanakorn C, Thongprasit S, Thamkongka S, Visoottiviseth V. ABIS: a prototype of android botnet identification system. In: *2016 Fifth ICT International Student Project Conference (ICT-ISPC)*. IEEE; 2016. p. 1–5.

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